

Title	Speaking into Dissonance: Foreign Language Learning, Epistemic Dissonance, and the Location of Difficulty
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Abstract

An accurate correction is not necessarily an adequate explanation. And an adequate explanation does not by itself mean that the person receiving it can evaluate it. This paper develops a normative account of explanatory authority: authority over where a difficulty is taken to lie, when it may have several sources and when those who receive explanations differ in their capacity to evaluate them. Second-language (L2) speaking is the principal case. When a speaker fails to say what she meant, the difficulty may lie in a slip, in missing knowledge, in the way her first language led her to frame what she meant, in the listener, or in the interaction, and several of these can contribute at once. L2 speaking is not unique in raising this question. It raises it recurrently, for ordinary people, and in a form that can make habitual framing observable. The paper uses an agent-relative concept of epistemic dissonance, an interval in which the location of a difficulty is unsettled for the speaker, to explain why the allocation of explanatory authority matters. That allocation governs whether such intervals can arise, how long they last, and what can be considered within them. The paper distinguishes legitimate closure and delegation from premature closure, foreclosure and abdication. It then proposes a criterion for teaching, listeners and AI language tools. Each attribution of a difficulty should keep within the scope its grounds support. Its grounds should be available for examination. And arrangements over time should develop, and then preserve, the learner's capacity to recognise, evaluate and, where warranted, contest how her difficulties are located. Evidence on the Foreign Language Effect is reviewed as context. It supports effects of reasoning in an L2 but not a lasting effect of learning one. The paper states testable hypotheses for the claims it cannot settle.

In Plain Language

You are speaking a foreign language and your sentence comes out wrong. Why? Perhaps you slipped, and you knew the right way to say it. Perhaps you never learned the word. Perhaps the other language simply does not say things the way your own language does, so what you were trying to say does not fit. Or perhaps the listener struggled with your accent and blamed you. Often it is more than one of these at once.

When someone corrects you, whether a teacher, a listener or an app, they also tell you, in effect, what went wrong. Usually they are right, and usually that helps. But sometimes they fix one real problem and leave the impression that it was the only one. Or the fix comes so quickly that you never get to wonder for yourself. Over time you can lose the habit of asking whether the problem lies in how you are looking at things, not just in a mistake you made.

This paper is about who gets to decide where a problem lies, and how to do that fairly. It does not say 'stop correcting people'. It says three things. Do not claim more than you know: 'that is the wrong verb' is fine, but do not make it sound like the whole story if it is not. Show your reasons, so the person can check and question them. And let people sometimes try to work out the problem

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themselves first, so that they learn to judge for themselves. Once they can, it is fine for them to rely on the teacher or the app.

This matters more now that translation and correction apps fix our language instantly, all day. They are very useful. But a tool that always fixes everything can quietly stop us noticing why something did not work. The paper suggests how such tools could be designed and judged so that they help people learn, not just do the work for them.

Note on This Version

Version 2.0 is a substantive revision. Version 1.0 argued that learning a foreign language trains tolerance of cognitive dissonance, and that this tolerance strengthens pluralist democracy. That argument depended on treating evidence about reasoning *in* a second language as evidence about the lasting effects of *having learned* one. It also contained several misdescriptions of its empirical sources. The present version withdraws the civic conclusion, changes the paper's central question, and adopts the concept of epistemic dissonance developed in Kahl (forthcoming), which is not defined by discomfort. Appendix A sets out the specific corrections to Version 1.0.

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About the Author

Peter Kahl is an independent researcher working on epistemic authority, answerability and the governance of knowledge in institutions and AI systems. He holds an LLB from Staffordshire University and an LLM from the University of Reading. His peer-reviewed article on autonomy and agency in AI systems appeared in *Discover Artificial Intelligence* in 2026.

The question this paper asks, who decides where a difficulty lies, is one he has met from several sides. He speaks six languages: Czech, his native language, together with English, German, Russian, Mandarin and Cantonese. He has also studied several others, including Italian, French and Spanish. He spent some twenty years working in the United States and about fourteen in Hong Kong, where he served as a Czech–English court interpreter. Interpreting in court is a setting in which a failure of understanding may lie in a word, in a witness's knowledge, or in the different ways two languages divide up an event, and in which it matters which. Before turning to law and philosophy, he spent about two decades in analogue semiconductor design and failure analysis, work organised around locating the source of a fault when several contributors are possible. At high school, he and his classmates, half in jest, devised an Esperanto-like constructed language, which earned a teacher's commendation.

About Lex et Ratio

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WORKING PAPER

Speaking into Dissonance

FOREIGN LANGUAGE LEARNING, EPISTEMIC DISSONANCE, AND THE LOCATION OF DIFFICULTY

Peter Kahl, independent researcher, Lex et Ratio
27 October 2026

1. Introduction

1.1 Opening vignette

You stand in front of people who speak the language natively and rehearse a sentence before letting it go. You know the words and you have studied the grammar. When you speak, the syllables stumble and your listeners' faces flicker: polite, puzzled, indulgent. You know something has gone wrong. You do not yet know what.

It may be a slip you would catch in a calmer moment. It may be a word or a construction you have never learned. Or the thing you were trying to say may not divide up in this language the way it divides up in your own. In that case part of what failed is not your performance but the way you composed the thought for speaking. These possibilities are not exclusive. A single failed sentence can involve all three, and at the moment of failure they are often indistinguishable.

1.2 The question

This paper asks who or what settles where such a difficulty lies, and with what consequences. A teacher may settle it by supplying the correct form, a listener by discounting what was said, a tool by replacing it. Often the settlement is right, and quick settlement is what learning requires. But a settlement can also be premature. It can assign the difficulty to the speaker before she has taken any part in placing it. Or it can identify one real contributor while concealing another. The concern is not discomfort, which this paper does not ask anyone to cultivate, but the allocation of authority over the explanation of a difficulty.

The paper is a contribution to social epistemology and the philosophy of education. L2 speaking is its principal case, chosen because it raises the question often and in an accessible form (Section 3.4). Teaching, listeners' credibility judgements and AI language tools are the three settings in which the question is examined. The paper draws on empirical work in psychology and applied linguistics. It

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does not claim to establish a new psychological effect, and where its claims depend on evidence that does not yet exist, it states them as hypotheses (Section 8).

Throughout, I use L1 for a speaker's first or dominant language and L2 for a second language acquired later. The terms are conventional in applied linguistics and imperfect, because dominance shifts and many multilingual people have no single first language. They are serviceable for the argument.

1.3 Two claims to keep apart

A body of experimental work, the Foreign Language Effect literature, finds that people judge differently, often with less emotional bias, when reasoning in an L2 (Keysar, Hayakawa and An 2012; Costa et al. 2014). It is tempting to infer that learning languages makes people more open-minded. That inference runs together two claims.

The *state claim* is that while a person reasons in an L2, her judgement differs in identifiable ways. The *trait claim* is that learning an L2 leaves a lasting change that persists when she reasons in her L1. Most of the evidence concerns the first. Only the second could support the familiar civic argument for language learning, because citizens mostly deliberate in their first language. Section 5 argues that the evidence for the trait claim is thin and in places points the other way. The argument of this paper does not depend on it.

1.4 Research questions

1. When does a communicative failure become an occasion of epistemic dissonance, as opposed to an occasion for correction?

2. Under what conditions does a mismatch between languages bear on a speaker's conceptual frame, as opposed to her linguistic encoding or pragmatic conventions?

3. What does reasoning in an L2 do to judgement, and does the evidence support any lasting effect of L2 learning?

4. In teaching, listening and AI-mediated language use, what distinguishes legitimate closure of a difficulty from foreclosure?

5. What criterion can be applied to these arrangements, and how could its consequences be tested?

1.5 Thesis

The paper's thesis is that the epistemic significance of a difficulty lies neither in its persistence nor in the learner's taking part in every correction. It lies in whether the allocation of explanatory authority keeps each placement within the scope its grounds support, and whether it preserves the learner's developing capacity to recognise, evaluate and, where warranted, contest how difficulties are located.

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Five supporting claims carry this thesis. First, epistemic dissonance is narrower than communicative failure. It requires that the explanation of a failure has registered for the speaker as unsettled, and many L2 failures do not meet that condition. Second, a linguistic mismatch does not by itself reveal a conceptual one. Language makes some of a speaker's framing habits observable while she speaks, but whether a mismatch bears on her frame in the relevant sense is a further question. Third, L2 speaking is not unique in raising frame-level questions. It is distinguished by how frequently and accessibly it raises them, and by how it makes implicit framing habits observable. Fourth, closure of a difficulty is often legitimate, and so is delegating its resolution. What is objectionable is closure that claims more than its grounds support, settlement that forecloses a question before it can arise when the question matters to the activity, and delegation that has become abdication. Fifth, these distinctions yield a criterion for teaching, listeners and AI tools, stated in Section 7.5.

1.6 Outline

Section 2 sets out the concept of epistemic dissonance and the conditions under which a failure becomes an occasion of it. Section 3 separates linguistic, pragmatic and conceptual mismatch and states the comparative claim about L2 speaking. Section 4 reviews what reasoning in an L2 does. Section 5 examines the state–trait gap. Section 6 turns to teaching, authority and listeners. Section 7 addresses AI-mediated language use and states the criterion. Section 8 sets out hypotheses and research designs. Section 9 draws implications, and Section 10 concludes.

2. Epistemic Dissonance and the Location of Difficulty

2.1 Three questions

Three questions are usually run together when a difficulty is corrected.

1. Was the correction accurate?
2. Was the explanation of the difficulty adequate?
3. Who had an effective opportunity to judge whether the explanation was adequate?

The answers come apart. An accurate correction can come with an incomplete explanation. An explanation can be intelligible without being exhaustive. And a learner can understand an explanation without having had any part in judging whether it captures her difficulty. Most evaluation of teaching, and almost all evaluation of AI language tools, concerns the first question. This paper concerns the second and third.

Throughout, a *placement* is an attribution of a difficulty to a proposed source. 'You used the wrong tense' places the difficulty in the speaker's performance. 'There is no single word for that in English' places it in the relation between two languages. *Explanatory authority* is control over which placements are offered, accepted and allowed to stand.

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2.2 From Festinger's drive

Festinger proposed that holding inconsistent cognitions produces psychological discomfort, which motivates a person to reduce the inconsistency by changing a belief, adding a justification, or avoiding information that would sharpen the conflict (Festinger 1957). In his framing, dissonance was a drive state, significant chiefly as something to be reduced. The best-known demonstration is the forced-compliance study, in which participants paid one dollar to tell another participant that a tedious task had been enjoyable later rated it more enjoyable than those paid twenty dollars (Festinger and Carlsmith 1959). Choice-induced preference change has also been reported in young children and capuchin monkeys (Egan, Santos and Bloom 2007). The method used in such studies can produce apparent preference change as a statistical artefact (Chen and Risen 2010), although studies designed to remove the artefact still find a reliable effect (Enisman, Shpitzer and Kleiman 2021). Nothing in the present argument depends on these findings.

2.3 Epistemic dissonance

This paper uses a concept that borrows Festinger's word and means something different. Following Kahl (forthcoming), *epistemic dissonance* is an agent-relative condition: something a person encounters does not sit within the frame presently organising her judgement, while the source or significance of the misfit remains unresolved *for her*. It is not defined by discomfort. An unresolved misfit can be registered with unease, with curiosity, or almost dispassionately. And it secures a possibility, never a mechanism. While the location of a difficulty is unsettled for her, a question becomes available that a clean placement does not raise: whether something in how she is framing the matter is contributing to it. Nothing compels her to ask it. Because the concept is agent-relative, it cannot be read from outside.

An explanation can be unsettled in two different senses.

- *Unsettled for the agent*: she does not know where the difficulty lies, although the evidence available to someone else, such as a teacher or a reference grammar, may settle it conclusively.
- *Underdetermined by the evidence*: what is available does not settle where the difficulty lies, for anyone.

Epistemic dissonance concerns the first. The two do not coincide. A learner may be unsure about a difficulty whose explanation her teacher holds on conclusive grounds. She may also feel certain about one that the evidence leaves open. The legitimacy of a placement therefore cannot depend only on whether the learner is unsure. It depends also on what its grounds support (Section 7.5).

Epistemic dissonance is not confined to communicative failure, or to language. A speaker can find her habitual categories inadequate while producing an utterance that is grammatical, appropriate and understood. The encounter is then conceptual, not communicative. And the concept, as developed in Kahl (forthcoming), applies wherever a competent frame meets material it does not readily hold. This paper examines one route by which it arises.

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2.4 Four conditions within communicative failure

Within communicative failure, four conditions can be distinguished.

1. *Communicative divergence*: intended meaning, produced expression and received interpretation fail to align. The divergence may lie between intention and production, between production and reception, or in the interaction between two systems that are each internally coherent. Whether something counts as a divergence can itself depend on the frame through which the exchange is assessed.

2. *Registered difficulty*: the speaker recognises that the exchange has not gone as expected.

3. *Unsettled explanation*: the speaker registers that she does not yet know where the difficulty lies.

4. *Frame-level consideration*: while the explanation is unsettled, the speaker considers whether the way she composed what she meant is among the contributors.

Conditions 1 to 3 form a progression: each presupposes the one before. Condition 4 is not simply a further step. It is a possibility that condition 3 opens. A speaker can also reach a frame-level *attribution* without passing through condition 3, as when a bilingual immediately recognises that a misunderstanding arises from different conventions of politeness. That is a settled frame-level explanation, not dissonance. And dissonance can arise, as Section 2.3 noted, without any communicative divergence at all. The four conditions describe one possible progression within communicative failure. They are not stages that every instance of dissonance passes through.

Epistemic dissonance, on this route, begins at condition 3. A learner who knows she has used the wrong verb ending is at condition 2: the location is settled for her. A learner who says something incorrect and at once assumes a slip, when the failure also reflects a mismatch between how her two languages divide up the matter, has registered a difficulty and settled its location without the mismatch coming into view. Nothing was open in her judgement, so nothing was closed in it. What was lost, if anything, was an occasion on which it might have opened. Section 2.6 names this difference.

2.5 Locus and type

The sources to which a difficulty can be attributed vary along two dimensions.

Dimension	Categories
Locus: where the contributor sits	The speaker; the listener; the interaction between them; the wider linguistic or institutional arrangement
Type: what kind of contributor it is	Performance error (knowledge held but misapplied); knowledge gap (knowledge lacking); framing mismatch (categories or habits of attention that do not fit the matter or the other party's)

A difficulty can be classified on both dimensions, and contributors can coexist. Consider an English speaker learning Mandarin who tries to express a temporal distinction using a construction modelled on English. Her failure may involve a performance error and a knowledge gap, both located in her,

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together with a framing mismatch located in the relation between her habitual English distinctions and those Mandarin grammaticalises. A listener can likewise err, lack knowledge or frame badly. A difficulty can also arise in the interaction when neither party has erred. Applied linguistics has long treated such cases as interacting phenomena, and nothing in the classification denies this.

Two consequences matter later. First, a correction can identify one real contributor correctly while concealing another. The difficulty is then not that the wrong source was chosen. It is that one correct source was presented as the whole explanation (Section 7). Second, a placement can itself be mislocated: a contributor at one locus can be attributed to another. The credibility penalty discussed in Section 6.5 has this form. A processing load at the listener's locus is attributed to the speaker's claim.

2.6 Closure, delegation and their failures

Closing a difficulty is often exactly right. The argument needs the distinctions summarised below. It does not need a general theory of epistemic closure, and it does not attempt one.

Term	Meaning	Normative status
Resolution	A difficulty receives an adequate explanation	Generally desirable
Closure	An explanation is accepted as sufficient for the present purpose	Often legitimate
Premature closure	A question the agent actually holds open is settled, by authority or convenience, beyond what the grounds support	Objectionable
Foreclosure	An arrangement settles in advance where difficulties must lie, or narrows the availability of a source, so that a question cannot arise or is costly to raise	Objectionable where the question matters to the activity's purpose
Delegation	The agent relies on another person or system to locate a difficulty	Often legitimate
Abdication	The agent no longer retains effective evaluative authority over what she has delegated	Objectionable

'Foreclosure' is used in the sense of Kahl (forthcoming). It names arrangements that fix the location of difficulty *before* the learner encounters anything, or that make a source practically unavailable. Premature closure acts on a question the learner already holds. The two can occur together but work differently. A tool that silently replaces every non-standard construction mostly forecloses: it prevents occasions of dissonance from arising, without closing any the learner had. A teacher who answers a learner's open question about why something 'cannot be said' with an authoritative 'it's just wrong' closes prematurely. Foreclosure need not be explicit. A classroom that answers every difficulty with a grammatical correction never says that grammar is the only possible explanation. Its repeated practice nonetheless establishes grammar as the only recognised kind.

Delegation and abdication are distinguished by what the agent retains, not by whether she judges personally. What 'effective evaluative authority' requires is developed in Section 6.3. The same distinction, between delegating a judgement and losing evaluative control over it, is developed for fiduciary and administrative authority in Kahl (2026). This paper applies it only to the location of difficulty.

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2.7 What dissonance does in the argument

The criterion in Section 7.5 applies whether or not a particular learner is in dissonance. A tool that gives accurate but misleadingly exhaustive explanations violates it even if no learner ever registers a difficulty. A classroom whose practice forecloses frame-level questions is objectionable even if no student has yet felt an unresolved misfit. Agent-relative dissonance cannot be detected from outside, so a criterion that applied only when it occurred could never be applied.

Dissonance is therefore not a condition of the criterion's *application*. It is the reason the criterion *matters*. The concepts relate as follows.

- *Epistemic dissonance* concerns the agent's condition: an interval in which the location of a difficulty is unsettled for her, so that her own frame can come into question.
- *Premature closure* concerns intervention in that condition once it exists.
- *Foreclosure* concerns the prior organisation of opportunities for that condition to arise.
- *Explanatory authority* concerns the distribution of control over all three.

Without the concept of dissonance, the scope requirement in Section 7.5 would be a generic norm of honest explanation, and there would be no account of why over-scoped placements and foreclosing practices are costly in a particular way. The concept supplies that account. Such placements and practices govern whether the interval in which a frame can come into question is able to arise or to last. That is why the allocation of explanatory authority has consequences beyond the accuracy of any single correction.

2.8 Relation to accounts of epistemic dependence and authority

The paper's account of delegation sits within a familiar philosophical position. Hardwig (1985) argued that a great deal of what people rationally believe rests not on evidence they possess but on trust in others who possess it, and that such dependence is a normal condition of knowledge, not a failure of it. Zagzebski (2012) developed an account of epistemic authority on which a conscientious person may rationally let an authority's belief stand in for her own reasoning in a domain, when she judges, on reflection, that doing so is more likely to yield beliefs that survive her own scrutiny. On Zagzebski's 'preemption' thesis, a legitimate authority's belief replaces, and does not merely add to, the believer's other reasons.

The present paper accepts both positions for what they cover, and adds two points. First, deferring to an authority about a *first-order* matter, such as which form is correct, need not extend to deferring about the *location* of one's difficulty. A learner can rightly accept that a form is wrong while the question of why her attempt failed remains hers. The authority that justifies accepting a correction does not by itself justify accepting the scope of the explanation that comes with it. Second, Zagzebski's justification of deference rests on the believer's own reflective judgement that deference is reliable. A learner still acquiring a domain may not yet be able to make that judgement about

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placements in it. The developmental structure of Section 6.3 follows from this. A fuller positioning of the argument within the literature on epistemic authority, dependence and intellectual autonomy is left for further work.

2.9 What the concept does not license

Four limits follow.

First, the concept gives no reason to manufacture difficulty or prolong uncertainty. A difficulty whose location is already established is not an occasion of dissonance, and settling it at once loses nothing.

Second, framing mismatch is not privileged. Most difficulties are errors or gaps. A speaker who attributes every difficulty to her frame judges no better than one who attributes every difficulty to her own error. The aim is appropriate discrimination between sources, not more frame-level attribution.

Third, openness has costs. Investigating alternative explanations takes time, attention and effort, and a learner may rationally delegate a judgement because the expected benefit of making it herself is small. Such delegation can be an exercise of epistemic agency, not a loss of it.

Fourth, because dissonance cannot be detected from outside, no teacher or tool can act on it directly. What arrangements can act on is whether the occasions they create pre-specify where a difficulty must lie (Kahl forthcoming). The criterion therefore applies to arrangements over time, not to single exchanges.

3. Language, Pragmatics and Concepts

3.1 Three levels of mismatch

A mismatch between two languages can occur at three levels, and they need to be kept apart.

Level	Nature of the mismatch	Example
Linguistic	Differences in lexical, grammatical or semantic encoding	Different tense and aspect systems; obligatory evidential marking in some languages and not others
Pragmatic	Differences in the conventions governing appropriate expression	Forms of address; politeness and register; how directly a refusal may be stated
Conceptual	Differences in the distinctions or assumptions through which a person interprets what she is talking about	Treating a classification that is culturally contingent as though it were simply how things are

Only the third level bears on a speaker's frame in the sense that matters for epistemic dissonance. It cannot be read off the first two. A language may oblige its speakers to mark a distinction without its speakers being the only ones able to think it. English, for example, has no grammatical system of evidentials, morphology that marks whether information was witnessed, inferred or reported. Yet English speakers can readily distinguish what they saw from what they were told. A speaker may likewise think distinctions her language does not grammaticalise. The absence of a word is not the

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absence of a concept. A linguistic mismatch may therefore reveal only a difference in encoding. It may reveal a pragmatic difference. Occasionally it may prompt a speaker to reconsider an assumption she had not noticed she held.

3.2 Thinking for speaking

This paper does not adopt linguistic determinism, the view that a language fixes the categories available to its speakers' thought. The position it needs is narrower, and it is well established in the study of language and cognition. Slobin's 'thinking for speaking' hypothesis holds that a language trains its speakers to attend to particular dimensions of experience *in the act of composing speech*, because the language obliges them to encode those dimensions (Slobin 1996). Speakers of a language that obligatorily marks aspect must attend to whether an event is complete in order to speak about it at all. The hypothesis concerns habits of attention during speech production. It does not claim that speakers cannot think what their language does not mark.

Research on crosslinguistic influence in second-language learning makes a compatible distinction. Jarvis and Pavlenko (2008) distinguish *semantic* transfer, carrying over L1 word meanings and their structure, from *conceptual* transfer, carrying over the L1-based conceptual categories through which the speaker represents the matter. They treat the two as different phenomena with different evidence.

Together these provide what the paper's argument requires. When an L2 speaker composes what she means, she composes it partly through L1 habits of attention: which dimensions she treats as needing specification, which she leaves unmarked, how she carves the event. When the L2 requires different specifications, those habits become observable, to her as well as to others. This is the sense in which a speaker's framing becomes visible in L2 speech. Her language is not her conceptual frame. But her L1 thinking-for-speaking habits are part of how she composes what she means, and the L2 exposes them.

3.3 When a linguistic mismatch becomes a frame-level question

A linguistic mismatch becomes a frame-level question only through a sequence of further recognitions. Each is harder to reach than the one before.

1. *Recognition of a linguistic difference*: the L2 encodes the matter differently. 'Mandarin marks aspect where English marks tense.'

2. *Recognition of a habitual practice*: the difference exposes something the speaker habitually attends to, or habitually leaves unmarked, when she composes speech. 'I always specify when, and rarely whether it was completed.'

3. *Recognition of possible influence*: that practice may be shaping how she represents the matter itself, not only how she says it. 'Perhaps I have been treating the timing of events as more basic than their completion.'

4. *Evaluation*: whether that influence produces a substantive distortion for the purpose at hand.

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The move from stage 2 to stage 3 is the move from linguistic awareness to conceptual reflection, and nothing about L2 speaking guarantees it. A speaker can learn that Mandarin requires a different expression of aspect without reconsidering her understanding of time or events, and usually she should not need to. What makes stage 3 possible is not the L2 itself but its combination with an occasion on which the difficulty's location is open. The speaker must not already have been told that the problem is simply a form to be learned. This is why the allocation of explanatory authority, discussed in Sections 6 and 7, bears on the frame-level location at all.

Stage 4 matters as much as stage 3. Not every habitual framing practice is defective, and different ways of dividing up a matter can be equally adequate for different purposes. The point of reaching stage 3 is not to abandon an inherited frame because another language encodes things differently. It is to recognise that some framing choices are contingent, and then to assess whether they serve the purpose. Most such assessments will conclude that the habit is harmless. The claim of this paper is therefore not that L2 speaking often produces conceptual revision. It is that L2 speaking often produces occasions on which stage 3 becomes possible, and that what happens on those occasions depends on how the difficulty is handled.

3.4 A comparative claim

It is tempting to treat the three-location structure as distinctive of L2 speech. It is not. The same structure appears wherever a competent frame meets material it does not readily hold. A physicist whose results conflict with an accepted model faces possible experimental error, missing information, or an inadequacy in the model. A lawyer working in an unfamiliar legal system may misapply a rule, lack knowledge of the jurisdiction, or assume wrongly that a familiar legal category has an equivalent there. Comparative law is an especially close analogue to L2 speaking. Its practitioners know that concepts which look equivalent can do different work within different institutional systems.

What L2 speaking offers is not a unique structure but three properties in combination.

- *Frequency*: mismatches arise constantly, in ordinary exchanges, not only at the frontier of expert work.
- *Accessibility*: they arise without specialised knowledge, for anyone learning a language.
- *Externalisation*: the L2's different requirements make ordinarily implicit features of the speaker's framing observable, because what she must specify differs from what she is used to specifying.

None of these properties is unique to L2 learning, and their comparative magnitude has not been measured. The defensible claim is therefore modest. L2 speaking is a recurrent and comparatively accessible setting in which ordinarily implicit features of a speaker's framing may become objects of attention. That makes it a good case for studying how the location of difficulty is allocated. It does not make it the only case, or necessarily the best.

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4. What Reasoning in a Second Language Does

This section and the next provide empirical context. The paper's normative argument does not depend on them, and they are kept brief.

4.1 The Foreign Language Effect

Framing effects in risky choice are reduced when people work in a foreign language (Keysar, Hayakawa and An 2012). People are more willing to make the utilitarian choice in moral dilemmas such as the footbridge case when the dilemma is presented in a foreign language (Costa et al. 2014). A meta-analysis concluded that the effect is reliable and was not moderated by L2 proficiency, age of L2 acquisition, type of decision problem or mode of presentation (Del Maschio et al. 2022). Two explanations are usually offered, and they are not exclusive. L2 processing is more effortful, and an L2 carries less emotional resonance. The second is consistent with the finding that taboo words and reprimands produce stronger skin-conductance responses in bilinguals' L1 than in their L2 (Harris, Ayçiçeği and Gleason 2003). These findings concern particular decision paradigms. They do not show that L2 reasoning is better reasoning in general, or that the outcomes reflect a single process.

4.2 Belief revision, and why more is not better

Kaplan, Gimbel and Harris (2016) scanned forty politically liberal adults reading counterarguments to strongly held beliefs. Political beliefs were revised less than non-political ones. Greater resistance was associated with more activity in the amygdala and insula, regions linked to emotion and its bodily signals. Miller et al. (2025) adapted the design to 70 English-speaking monolinguals and 133 English–Spanish bilinguals, of whom 67 completed the tasks in English (L1) and 66 in Spanish (L2). All were politically liberal and mostly US-based. Bilinguals revised more in the L2 than in the L1, and bilinguals overall revised more than monolinguals. L2 processing slowed reading and rating, and slower times went with greater revision. Among L2 users, those reporting less emotional connection to Spanish revised most. Proficiency did not predict revision.

Greater belief revision is not, by itself, epistemic improvement. A person may revise because she is responding appropriately to evidence. She may also revise because the original belief is less salient in the L2, because she understood the counterarguments less well, or because she is less able to evaluate them. Every participant in Miller et al. met counterarguments chosen to be persuasive, so the design cannot distinguish these. What would show an epistemic benefit is not a greater *magnitude* of revision but greater *sensitivity to evidential quality*. Hypothesis H1 is framed accordingly.

4.3 Tolerance of ambiguity and neural evidence

Bilinguals answering a tolerance-of-ambiguity questionnaire in English (L2) scored somewhat higher than those answering in Italian (L1). The effect was small, concentrated in items about challenging perspectives and change, and measured by self-report (Purpuri et al. 2024). Neural evidence is limited. He et al. (2021) found that in a gambling task with feedback in Chinese (L1) or English (L2),

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loss aversion in the L2 context was mediated by the dorsolateral prefrontal cortex, and positive L2 feedback was followed by more gambling. This shows that emotional feedback is processed differently across languages. It is not evidence of a general damping of emotion in L2 use, and it does not concern belief revision. Because most brain regions serve many functions, I treat neural findings as consistent with the behavioural ones, not as independent confirmation of a mechanism.

4.4 Two affective channels as competing pressures

The Foreign Language Effect says an L2 *mutes* emotion. Yet L2 speaking is often experienced as *heightened* emotion. Both can be true because they concern different things. The Foreign Language Effect concerns the emotional resonance of *content*. The emotions of L2 speaking concern *performance*. Horwitz, Horwitz and Cope (1986) identified foreign language classroom anxiety as a distinct form of anxiety, involving communication apprehension, fear of negative evaluation and test anxiety. It is about the saying, not what is said.

I suggest the two channels create competing pressures on how a difficulty's location is handled. Distance from content may make it easier to treat what one is saying as revisable. Exposure in performance may push a speaker to settle a difficulty quickly, and in herself. This is a hypothesis, and neither channel has a fixed epistemic direction. Emotional distance may reduce engagement; performance anxiety may prompt careful self-monitoring. Their effects probably depend on task, setting, proficiency and the individual, and they can be tested by manipulating the two separately (H5). Kahneman's (2011) contrast between fast, intuitive and slow, deliberative processing is a useful shorthand for the effortfulness explanation, but it is a heuristic, not a mechanism.

5. State and Trait

5.1 The inference and the evidence

The familiar civic argument for language learning, in the form Version 1.0 of this paper gave it, requires that L2 learning produce a lasting disposition that carries over to reasoning in the L1. The evidence in Section 4 concerns L2 *use*. Neither within-person L1–L2 comparisons nor comparisons with monolinguals can establish a trait effect without addressing selection into language learning and confounding.

The strongest support for a lasting association is correlational. In a survey of 2,158 people, multilinguals scored higher than monolinguals and bilinguals on tolerance of ambiguity, higher overall proficiency went with higher tolerance, and time abroad was associated with higher tolerance, levelling off after about a year. The authors note that the relationship may run both ways (Dewaele and Li 2013). Other findings cut against a training interpretation:

- years of L2 study had no significant effect on tolerance of ambiguity (Purpuri et al. 2024);

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- proficiency did not predict belief revision, while weaker emotional connection to the L2 did (Miller et al. 2025);
- in a pseudo-longitudinal study of 189 pupils, foreign language classroom anxiety varied little between ages 12 and 18 (Dewaele and Dewaele 2017), which is not what simple habituation would predict, though a cross-sectional design cannot settle it;
- after correction for publication bias, there is no systematic bilingual advantage in adult executive functioning (Lehtonen et al. 2018).

5.2 The distance hypothesis

One developmental hypothesis deserves statement. If greater revision in an L2 depends on emotional distance from it, then living in the language, forming relationships in it and coming to feel in it might reduce the distance, and with it the effect. The evidence does not establish this. That emotional connection predicts revision while proficiency does not shows neither that connection grows with learning nor that growing connection weakens the effect. Proficiency and emotional integration vary separately. The test would be longitudinal: following learners as the L2 becomes a language of intimate relationships, professional identity or everyday emotional life, and asking whether changes in attachment predict changes in the effect independently of proficiency.

5.3 What survives

Nothing in the rest of the paper depends on a trait effect. What survives is a claim about L2 speaking as a *practice*: a recurrent setting in which communicative difficulties sometimes register with their explanation unsettled (condition 3), and in which the frame-level location is sometimes among the live candidates. Whether repeated encounters with such occasions affect judgement elsewhere is an open question, which H3 is designed to test. The normative argument does not wait on the answer. It concerns how the occasions themselves are handled.

6. Who Locates the Difficulty: Teaching, Authority and Listeners

6.1 Correction, closure and foreclosure

Every practice of instruction requires learners to accept conventions they did not invent, and language learning more than most. Conformity to grammatical convention is not epistemically suspect, and correction is not either. When a learner misconjugates a verb and the teacher supplies the correct form, nothing need have been lost. The difficulty had a known address, and asking the learner to investigate alternative explanations would be inefficient and confusing. The same holds where the learner herself is unsure but the teacher holds conclusive grounds. The difficulty was

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unsettled *for her*, not underdetermined, and a correction that says what it covers closes it legitimately (Section 2.3).

The concern arises in two narrower ways, corresponding to the distinction drawn in Section 2.6.

Premature closure occurs when a learner holds a difficulty open and a placement settles it beyond what its grounds support. A learner asks why something she wants to say seems impossible in the L2. She is told that her sentence is simply wrong, when the grounds support only the claim that the form is ungrammatical, and say nothing about whether what she meant can be said.

Foreclosure, in the sense of Kahl (forthcoming), is an arrangement that settles in advance where a learner's difficulties must lie, in her application, knowledge or reasoning, so that a frame-level question cannot arise whatever she meets, or can be raised only at a cost. Foreclosure is distinct from structure. Teaching established results as established forecloses nothing, and most instruction rightly arrives with its difficulties already located (Kahl forthcoming). The objection is to arrangements in which no occasion leaves the location open, where the activity's purpose includes the learner's coming to judge for herself.

Language teaching has long recognised the relevant variation in practice. In a study of four French immersion classrooms, Lyster and Ranta (1997) distinguished *recasts*, in which the teacher reformulates the learner's utterance correctly, from moves that withhold the correct form and prompt the learner to repair her own utterance: elicitation, metalinguistic clues, clarification requests and repetition of the error. Teachers relied overwhelmingly on recasts, but recasts were the least likely to lead to learner-generated repair.

The present framework adds something to this tradition but does not replace it. The corrective-feedback literature asks which moves best produce accurate forms and uptake. The question here is who holds authority over the *explanation* of a difficulty. A recast settles the location and supplies the answer. A prompt usually still settles the location, signalling that there is an error and roughly where, while leaving the repair to the learner. Neither, as usually practised, asks the learner what kind of difficulty she thinks she had. That further move is not needed for most errors and should not replace correction. Its value lies in its being available on some occasions.

The second affective channel (Section 4.4) bears on this. A learner exposed in front of peers has reason to accept the quickest placement on offer, because it ends the exposure. If the competing-pressures hypothesis is right, the occasions on which a learner most needs a share in placing a difficulty are those on which she is least likely to claim it.

6.2 Cultural variation

Whether accepting or contesting a placement expresses deference, belonging or independence varies between cultures. Güngör et al. (2014) surveyed 163 Japanese and 172 Turkish students. Both cultures emphasise interdependence, but differently. Japanese participants described their agency more in terms of conformity, or fitting in. Turkish participants described it more in terms of

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relatedness, or sticking together. Conformity predicted relational well-being in the Japanese sample, and relatedness did so in the Turkish sample. Autonomy predicted personal well-being in both.

A learner's willingness to contest a teacher's placement of her difficulty is therefore not a simple measure of critical independence. A student who accepts correction without demur may be maintaining a relationship that matters to her well-being. A student urged to 'speak up' may be placed under a pressure her own norms do not recognise. What the argument requires does not depend on any single ideal of the autonomous, challenging learner. It requires that the arrangement leave room for the frame-level possibility, whatever the register in which a learner then engages with it.

6.3 Instructional dependence, delegation and clientelism

Epistemic Clientelism Theory holds that knowledge practices are structured by asymmetries of authority, and that people defer to epistemic patrons in exchange for recognition, security or advancement (Kahl 2025b). In a clientelist relation the patron supplies not only knowledge but placements. The client receives the relief of a settled location and gives up her share in settling it.

Dependence on expert correction is not, by itself, clientelism. A responsible teacher necessarily exercises asymmetrical authority, and dependence on others is a normal condition of knowledge (Hardwig 1985). Three markers distinguish legitimate dependence from the self-reinforcing kind.

- *Contestability*: a placement can be questioned without penalty.
- *Available reasons*: the grounds of a placement can be given and examined, not only its result.
- *Growing capacity*: the arrangement transfers the placing of difficulties to the learner as her competence grows.

What effective evaluative authority requires. The difference between delegation and abdication turns on whether the learner retains *effective* evaluative authority over what she has delegated. Three senses of authority need distinguishing, because they can come apart.

- *Formal authority*: the learner is permitted to reject a placement.
- *Practical authority*: she has the knowledge, time and resources to evaluate it.
- *Epistemic authority*: she can recognise when evaluation is warranted and make an adequately informed judgement when it is.

Formal authority alone is not enough. A tool may let its user reject every correction, while presenting its placements with such fluency and confidence that she cannot tell when rejection would be appropriate. Contestability then exists on paper, but effective evaluative authority does not. Delegation remains delegation only where practical and epistemic authority are retained as well as formal authority. This is the same distinction between holding authority and holding evaluative control that Kahl (2026) draws for delegated discretion in fiduciary and administrative law.

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Task-relative participation. It is tempting to say that beginners need to take part in placing their difficulties while advanced learners may delegate. That is too simple, because competence is specific to a domain and a task. A learner can be expert in grammar and a novice in pragmatic convention. An advanced learner may rationally delegate routine grammatical correction while meeting subtler pragmatic or conceptual difficulties that her competence is not yet equipped to evaluate. The more robust principle is that a learner's participation in placing difficulties should be concentrated where her evaluative competence is inadequate *relative to the difficulty* in question. For a beginner that covers most difficulties. For an advanced learner it may cover few, but those few may be exactly the ones where an authoritative placement is most likely to be over-scoped. The objectionable case is an arrangement that offers delegation for a class of difficulty before the learner can evaluate placements in that class, so that delegation is abdication from the start.

A limit of retained authority. Even effective evaluative authority protects better against some failures than others. A learner who intervenes 'when something unusual occurs' is well protected against routine errors. She is less well protected against concealment. A placement that presents one real contributor as the whole explanation does not look unusual, and that is why the other contributor stays hidden. Retained evaluative authority is therefore necessary but not sufficient. It must be paired with placements that do not claim more than their grounds support (Section 7.5).

Where the three markers are absent and the absence persists, the relation has become clientelist in the sense that matters: placements cease to be provisional, and the learner comes to expect that difficulties will be located for her. A preference for having difficulties settled by authority can become a stable disposition where deviation is costly (Feldman 2003). That is a reason to attend to how arrangements distribute the cost of leaving a difficulty unlocated. It is not a claim that language classrooms produce authoritarian dispositions.

6.4 A classroom case

Kurnianto (2021) describes a single teacher preparing a mixed-ability class at the Indonesian Civil Aviation Institute for the TOEFL English test, with a required score of 500. The results section reports 21 students; the abstract and conclusion give 24, and I follow the results. On the pre-test, 81% scored below 400 and 19% between 400 and 500. The teacher used cooperative learning with flexible grouping. At first groups formed by seating, so slower learners tended to sit together. Later the teacher let students arrange their own groups, and slower learners more often joined faster ones. The teacher also switched frequently to Indonesian to explain material. On the post-test, 57% scored above 500, 38% between 400 and 500, and 5% below 400.

The study documents teaching practices and test outcomes. It does not examine how learners understood or classified their difficulties, and nothing in it shows that they did so. The case is relevant here only for what it makes possible. Permitting discussion in the L1 *could* enable learners to articulate difficulties that their L2 competence does not yet let them explain, including what kind of difficulty they think they are having. That is a precondition for distinguishing an error from a gap from a framing mismatch. An L2-only rule would make that distinction hardest to draw exactly where

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it is most needed. Whether L1 discussion in fact has this effect is an empirical question that the case does not answer.

6.5 Accent and credibility

The location of a difficulty is also settled by listeners. Lev-Ari and Keysar (2010) had American participants rate the truth of trivia statements read aloud by native speakers and by non-native speakers with mild or heavy accents. Participants were told that the speakers were reading statements supplied by the experimenters, not stating their own knowledge. Statements were nonetheless rated as less true when spoken with an accent. The authors attribute this to processing fluency: listeners mistake the difficulty of processing accented speech for doubt about its content. When participants were told the study concerned accent and credibility, the penalty for mild accents disappeared, but the penalty for heavy accents remained.

In the present framework this is a clear case of a difficulty placed in the wrong location. A processing load at the listener's locus (Section 2.5) is read as a defect in the speaker's claim. Its relation to testimonial injustice needs care. Fricker (2007) defines testimonial injustice as a credibility deficit a speaker suffers because of identity prejudice in the hearer. The observed penalty is consistent with the processing-fluency mechanism Lev-Ari and Keysar propose, and their design reduced some routes by which prejudice could operate. It cannot exclude all of them. Accent is a social marker, and a fluency-based penalty can be embedded in, or amplified by, prejudicial judgements about who the speaker is. Where it is, testimonial injustice in Fricker's sense may be present. The finding does not by itself establish it, and it does not establish its absence.

The distinction is worth keeping because the residue is interesting. A credibility distortion can be epistemically consequential, and fall systematically on a social group, without demonstrable prejudice. The L2 speaker then faces a placement of difficulty in herself that she cannot easily contest, because contesting it requires the credibility that has been discounted. Accounts of epistemic injustice that do not rest on an identity-prejudice condition may be better placed to describe such distortions. Engaging them is a task for further work.

7. AI-Mediated Language Use

7.1 What the tools do

AI tools now intervene in L2 use at several points. Machine translation renders what a user composed in one language into another. Correction and completion tools detect deviations from expected forms and offer replacements, often before the user has finished. Conversational agents give the learner a partner for speaking practice. These are generative AI chatbots that converse by text or voice, sometimes with an animated face. Such agents can score, evaluate and remember. What distinguishes them is that they do not necessarily impose the social-evaluative pressures of a human listener. Each

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kind of tool intervenes at a different point in the structure set out above. Each can be assessed by the same question: what does it do to the allocation of authority over where a difficulty lies?

7.2 Evidence on speaking anxiety

The clearest evidence concerns conversational agents and performance anxiety. Wang et al. (2024) compared three groups of 33 Chinese undergraduate learners of English. A control group practised speaking with the teacher and classmates, one group used a text-and-voice chatbot, and one used a chatbot with a human-like avatar. Compared with the control group, the avatar group showed significant gains in willingness to communicate and in self-perceived communicative competence, and a reduction in speaking anxiety. Speaking performance did not differ significantly between groups. Ding and Yusof (2025) compared 60 students preparing for the IELTS English test, half of whom practised for six weeks on an AI conversation platform. That group improved more in speaking scores and showed a greater reduction in speaking anxiety. Both studies are small and short, and neither measured how learners located their difficulties.

Lower anxiety is relevant to the present argument but does not settle it. A tool that lowers exposure removes one pressure to settle a difficulty quickly and in oneself. It may therefore make occasions of open location more likely. It is equally compatible with a tool that resolves each difficulty so smoothly that its explanation never registers as open at all.

7.3 Substituting judgement and supporting it

A simple contrast between tools that resolve difficulties and tools that flag them is too coarse. A tool may:

- correct an error without explanation;
- correct an error and explain its cause;
- flag an error without identifying its cause;
- offer several alternative expressions;
- identify a possible cross-linguistic mismatch;
- ask the learner what she meant to say;
- delay correction until the learner has attempted her own diagnosis.

These are different forms of intervention, and their epistemic effects do not line up neatly. An immediate correction with a clear explanation may leave the learner more able to understand her difficulty than an unexplained flag. Repeated flags without adequate guidance may produce frustration without understanding. The relevant distinction is between tools that *substitute* for the learner's judgement about where a difficulty lies and tools that *support* it.

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Coexisting locations (Section 2.5) sharpen the point. A correction system may correctly identify a grammatical error while the failure also reflected a mismatch in how the learner's two languages carve the matter. Its correction is then accurate, and may even be well explained, while presenting one real contributor as the whole explanation. Machine translation shows the same pattern in a stronger form. A system renders whatever the user composed in whatever form it judges closest. A distinction the target language lacks is silently dropped, and one it requires is silently supplied. The user may never learn that a mismatch occurred.

7.4 Offloading and desirable difficulty

Two established literatures bear on this. Research on cognitive offloading studies how people use external aids to reduce the cognitive demands of a task, and how this affects what they themselves retain (Risko and Gilbert 2016). Offloading is often rational, but what is offloaded is not practised. Research on desirable difficulties finds that conditions which make learning feel harder and slower, such as spacing practice or generating an answer instead of reading it, often improve long-term retention, though learners tend to prefer easier conditions (Bjork and Bjork 2011). A more recent and contested strand concerns generative AI directly. In a preprint study of 54 participants writing essays with a large language model (LLM), a search engine or no tool, Kosmyna et al. (2025) reported weaker brain connectivity, measured by electroencephalography (EEG), in the LLM group. LLM users also reported less ownership of their essays. The study has not been peer-reviewed, its sample is small, and its neural measures admit more than one interpretation.

The present framework identifies a specific thing that may be offloaded. It is not only the work of finding the right form but the judgement of where the trouble lay. That judgement is easy to overlook because a substituting tool's output conceals that it was made.

7.5 The criterion

The argument yields a criterion that applies to teaching arrangements, to listeners' practices where these can be shaped, and to AI tools. It has three parts.

1. Scope. Each placement should keep within the scope its grounds support, and present itself as covering that scope. This replaces a requirement of non-exhaustiveness, which would be too demanding. A placement can be complete, and can rightly say so, relative to the question its grounds answer. 'This sentence is ungrammatical because this verb requires that construction' may be a complete account of the grammar. What it must not do is present itself as a complete account of why the communication failed, unless its grounds reach that far. Scope-indexing does not require hedging. A tool with adequate grounds to identify a straightforward grammatical error should not routinely add that other explanations might exist. It requires only that the claim match the grounds.

Scope-indexing does not escape the limits of the judgement that applies it. Deciding what one's grounds establish is also done through one's own frame. What it does is reduce the burden, and the reduction rests on an asymmetry. Judging that one's grounds positively establish something, that this verb requires that construction, is a bounded judgement about evidence one holds. Judging that no

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other contributor exists is a judgement about everything one's frame might be missing, and a competent frame is exactly what does not volunteer itself as a source of difficulty (Kahl forthcoming). A requirement that each placement be warranted *as complete* would rest on the second judgement. A requirement that each placement claim no more than its grounds cover rests only on the first. The person correcting need not know whether a framing mismatch is present. She need only not claim to have ruled one out.

The scope requirement applies to what an arrangement communicates through repeated practice, not only to what a single placement says. A classroom or tool that meets every difficulty with a grammatical correction makes no explicit claim that grammar is the only possible source. If no other kind of source is ever offered or acknowledged, its practice nevertheless communicates that claim. Over-scoping by repetition is a form of foreclosure (Section 2.6), and part 3 below is the remedy for it.

The scope requirement is applied to the state of the evidence, not to the learner's uncertainty (Section 2.3). Where the evidence settles a difficulty conclusively, a correction can close it, however unsure the learner was. Where the evidence leaves the source underdetermined, a placement should say so. A tool that meets the criterion will sometimes have to state that it cannot tell whether a difficulty is grammatical, pragmatic or conceptual, and that meets the requirement better than a confidently manufactured explanation.

2. Grounds. The grounds of a placement should be available for examination and contest. This is necessary but not sufficient. A placement can be explained and contestable and still conceal a coexisting contributor, because a learner can contest only what she is shown. That is why part 1 is needed. For AI tools, the relevant educational requirement is *justificatory adequacy*: the tool should be able to give reasons that genuinely support its placement and that are accurate, relevant and appropriately scoped. It is not *process transparency*, a faithful reconstruction of how the tool produced its output. A language model's internal computation may not correspond to any human-interpretable sequence of reasons, and a correct explanation of a grammatical rule can be pedagogically sound without describing that computation. Claims about a tool's *own reasoning* are a separate matter. Research on chain-of-thought explanations in large language models has found that stated reasoning can omit factors that in fact drove the output (Turpin et al. 2023). A tool should therefore not present a justification as a disclosure of how it reached its placement, and users and institutions should not rely on it as one. This finding concerns explanations that purport to report a model's reasoning. It does not show that pedagogical explanations generated by such tools are generally unreliable.

3. Developing capacity. Over time, the arrangement should develop, and then preserve, the learner's capacity to recognise, evaluate and, where warranted, contest how difficulties are located. Where her evaluative competence is inadequate relative to a class of difficulty, this requires some occasions on which the source of a difficulty in that class is not supplied in advance, so that she places it herself *before* a placement is offered (Kahl forthcoming). It also requires that the kinds of source an arrangement acknowledges are not, through repeated practice, reduced to one. Where her competence

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is adequate, it requires that delegation remain delegation: that she retain formal, practical and epistemic authority over what she has handed over (Section 6.3).

Two qualifications apply. First, the criterion is purpose-relative. A person using translation software to obtain urgent medical help should not be asked to investigate cross-linguistic mismatches, and foreclosure is objectionable only where the question it prevents matters to the activity's purpose. The criterion applies to tools and arrangements whose purpose includes learning, and its demands grow with that purpose. Second, it operates under technical and epistemic constraints. Reliably identifying conceptual mismatches between languages is difficult. A system may produce a plausible explanation of a linguistic difference that misrepresents the semantic or pragmatic distinction involved. Some difficulties are genuinely underdetermined by the available context. And keeping questions open has costs in time and attention that a learner may reasonably decline to pay (Section 2.9).

The criterion overlaps with familiar requirements in AI governance, such as explainability, contestability and human oversight, but it does not reduce to them. Part 2 corresponds to explainability and contestability, specified as justificatory adequacy. Parts 1 and 3 have no ready counterpart. They address the case those requirements miss: an accurate, justified, contestable placement that presents one contributor as the whole explanation, repeated until no other kind of source is recognised, and delivered to a learner who has not yet acquired the capacity to notice.

8. Hypotheses and Research Designs

The hypotheses differ in how directly they test the paper's argument. H3 and H4 test it most directly. H1, H2 and H5 concern the surrounding psychological claims.

H1 (evidence sensitivity). Reasoning in an L2 changes the relationship between the strength of a counterargument and the degree of belief revision it produces. *Design:* bilingual participants meet strong and weak counterarguments, in their L1 and L2. *Measures:* four quantities must be kept apart. They are (a) evidential strength, as independently assessed; (b) the participant's comprehension of each argument, tested directly; (c) the participant's own assessment of its strength; and (d) actual revision. Translated materials must be checked for equivalence of strength across languages, not only of meaning. *Outcome:* the difference in revision between strong and weak counterarguments, compared across languages, conditional on comprehension. An epistemic benefit would appear as better discrimination in the L2 at equal comprehension. Greater revision across the board would indicate greater susceptibility to persuasion, or weaker understanding, not greater openness to evidence. Felt emotional distance should be measured as a moderator.

H2 (trait transfer). The extent of L2 learning predicts evidence-sensitive belief revision in *L1* tasks, after controlling for baseline openness to experience and for selection into language learning. *Design:* longitudinal or intervention studies; cross-sectional comparison cannot test it. This is the prediction on which the civic argument of Version 1.0 depended.

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H3 (explanatory judgement). Learners whose instruction regularly includes occasions on which they place a difficulty themselves, before a placement is offered, will show better explanatory judgement on novel, non-linguistic problems. *Design:* instruction must not teach any error–gap–framing terminology, so that the outcome measures a capacity and not learned vocabulary. Two kinds of outcome task are used. *Designed tasks* have a principal contributor constructed by the researchers, some designed as errors, some as gaps and some as framing mismatches, all unlabelled. *Open tasks* are ones in which several explanations are genuinely defensible. *Measures:* on designed tasks, the accuracy of participants' spontaneous explanations against the designed contributor, with coexisting contributors scored as secondary identifications. On open tasks: whether participants identify relevant alternatives, whether they distinguish established from speculative explanations, whether they recognise underdetermination, and whether they revise their diagnosis appropriately when new evidence is introduced. More frame-level attribution is not the target, and over-attribution to framing counts as error.

The open tasks address a reflexive difficulty. In designed tasks the researcher decides in advance where each difficulty lies, which is a form of the advance settlement this paper examines. Open tasks do not escape researcher authority altogether, because someone must still decide which explanations count as defensible. Three methodological requirements follow:

- explicit scoring criteria, fixed and published before data collection;
- independent adjudication of participants' explanations, by raters blind to condition;
- a procedure for accepting well-supported explanations the researchers did not anticipate, with the criteria revised and the revision reported.

Without these, the study would risk rewarding agreement with the researchers' own frame, which is the outcome the paper's thesis warns against.

H4 (tool design). Among learners using AI language tools for learning, stating an initial diagnosis of a difficulty before the tool's assessment improves later explanatory judgement beyond what is explained by effort, feedback or explanation practice. *Design:* all groups receive equivalent explanations, at the same point in each task, and spend matched time on it. Groups differ only in what they do before the tool's assessment appears.

- *Diagnosis group*: states what kind of difficulty she thinks she had.
- *Generation control*: produces an attempted correction of equal effort, without diagnosis. This separates diagnosis from ordinary generation effects, the benefit of producing an answer oneself (Bjork and Bjork 2011).
- *Explanation control*: is shown a correction and explains why it is appropriate, without determining where the original difficulty lay. This separates judgement about the location of difficulty from general metacognitive and explanatory practice.

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Measures: the H3 outcome measures, at immediate, delayed-retention and transfer test points, together with willingness to communicate, speaking anxiety and speaking performance. Transfer tasks must differ substantially from the training tasks in domain and format, so that a positive result reflects a capacity that extends beyond the training context and not near transfer. Delayed and far-transfer measures matter most.

H5 (competing affective pressures). Emotional resonance of content and anxiety about performance have separable effects on how quickly, and where, speakers settle the location of a difficulty. *Design:* manipulate the two independently, varying content resonance by the emotional loading of the material and performance anxiety by the presence of an evaluating audience. *Measures:* time to settlement and the source chosen.

9. Implications

9.1 Teaching

The implication for language teaching is not to correct less. Most difficulties have a known address and should be corrected promptly. It is to include some occasions on which the learner is asked what kind of difficulty she thinks she had before any placement is offered. A learner's report that something 'cannot be said' in the L2 can be treated as a claim worth examining at the linguistic, pragmatic and conceptual levels, not only as a gap to be filled. And the L1 can be permitted for discussing difficulties, even where the L2 is required for practising them. These practices extend the corrective-feedback tradition (Section 6.1); they do not replace it.

9.2 Higher education

Many disciplines depend on recognising when a difficulty lies in the governing frame, not in the data or the method, and on resisting a competent frame's tendency to place every anomaly elsewhere (Kahl forthcoming). Many ordinary educational activities can expose framing assumptions: history, literature, mathematics, scientific modelling, translation and comparative social inquiry among them. L2 learning is one such setting. It provides recurrent opportunities for noticing habitual linguistic and conceptual assumptions, under conditions whose stakes vary greatly. For a student in a classroom, a failed sentence costs little. For a migrant at a border, a witness in court or a professional under assessment, the same failure can carry serious social, legal or material consequences. Most linguistic mismatches, moreover, never become frame-level questions (Section 3.3). Nothing here shows that L2 learning is the best route to frame-level judgement, or that it routinely produces it. Whether the practice transfers to disciplinary judgement is the question H3 is designed to test, and the answer should not be presupposed.

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9.3 Procurement and governance of AI tools

As AI language tools become the ordinary medium of L2 use, the criterion in Section 7.5 becomes a practical question for institutions that adopt such tools for learning, for teachers who prescribe them and, potentially, for regulators of AI in education. Each can ask the same questions. Can the tool flag a difficulty without resolving it, and can the learner choose? Does it distinguish kinds of difficulty, and does it acknowledge when it cannot? Does it present a correction as the whole explanation? Do its patterns of use leave learners occasions to place difficulties themselves? These questions are answerable, and at present they are rarely asked.

10. Conclusion

This paper has argued that an accurate answer is not necessarily an adequate explanation. It has also argued that an adequate explanation does not by itself mean the person receiving it can evaluate it. Between them these claims define a form of epistemic authority, authority over where a difficulty is taken to lie, whose exercise has consequences beyond the accuracy of any single correction.

Epistemic dissonance explains why. It is the agent-relative interval in which the location of a difficulty is unsettled, so that the frame through which a person judges can itself come into question. Arrangements that over-scope their placements, close open questions prematurely, or foreclose questions before they arise govern whether that interval can arise or last.

Second-language speaking is a good case for the argument, though not the only one. It offers recurrent occasions, for ordinary people, on which habits of framing can become observable.

From these distinctions the paper derives a criterion with three parts. Each placement should keep within the scope its grounds support. Its grounds should be available for examination, as genuine justification, not as a claim to disclose how a tool reached its result. And arrangements over time should develop, and then preserve, the learner's capacity to recognise, evaluate and contest how her difficulties are located, with her participation concentrated where her competence is not yet adequate to the difficulty.

The limits are real. The empirical base for the psychological claims is narrow. Epistemic dissonance cannot be measured directly, so H3 and H4 measure explanatory judgement, not how a difficulty registered. The comparative claim about L2 speaking has not been quantified against other settings. And positioning the account fully within the literature on epistemic authority, dependence and autonomy remains to be done.

The paper began with a speaker who knows something has gone wrong but not where. What it asks of the teacher, the listener and the tool is not that they leave her in difficulty. When they tell her where the difficulty lies, they should claim no more than their grounds support. They should sometimes let her try first to say where she thinks it lies. And they should leave her, in time, able to tell when to ask. The difficulty need not lie only where the first authoritative explanation places it. It may lie

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partly with the listener, in the exchange between them, or in the language she brought with her. None of these possibilities is guaranteed to be considered, and each is easily closed. They are worth keeping open.

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Appendix A: Corrections to Version 1.0

The following statements in Version 1.0 misdescribed their sources, and are corrected in the present text.

1. *Güngör et al. (2014).* Version 1.0 described a comparison of Turkish and American adolescents. The study surveyed Japanese and Turkish students and found that conformity predicted relational well-being in the Japanese sample, and relatedness in the Turkish sample (Section 6.2).

2. *Kaplan, Gimbel and Harris (2016).* Version 1.0 stated that defending political beliefs activates the orbitofrontal and dorsolateral prefrontal cortex. These regions were more active for challenges to non-political beliefs and are linked by the authors to flexibility. Resistance was associated with the amygdala and insula (Section 4.2).

3. *He et al. (2021).* Version 1.0 stated that this study reports reduced amygdala activation in L2 reasoning. It does not. It reports dorsolateral prefrontal involvement in loss aversion in the L2 context of a gambling task (Section 4.3).

4. *Miller et al. (2025).* Version 1.0 described the comparison inconsistently, in one place against monolinguals only and in another against monolinguals and L1 bilinguals, and drew a trait conclusion from a state finding. The study compares L1 and L2 conditions within one bilingual population and against monolinguals, and proficiency did not predict revision (Sections 4.2 and 5.1).

5. *Brock et al. (2025).* Version 1.0 stated that partial learning is 'often squeezed out' as conformity intensifies. In the model, partial learning remains a stable outcome and can grow with conformity. The model is no longer discussed in the main text.

6. *Kurnianto (2021).* Version 1.0 reported 21 students without noting that the abstract gives 24, and framed the case in terms of conformity incentives, an interpretation the article does not itself offer (Section 6.4).

7. *Festinger (1957).* Version 1.0 attributed the one-dollar/twenty-dollar study to Festinger's 1957 book. It was reported in Festinger and Carlsmith (1959) (Section 2.2).

8. *Unsourced claims.* Version 1.0 stated without citation that shame is the most common affect reported by L2 learners, and that taboo words elicit weaker autonomic responses in L2. The first is withdrawn; the second is now supported by Harris, Ayçiçeği and Gleason (2003) (Section 4.1).

Version 1.0 also cited several of the author's works as unpublished manuscripts or with non-DOI links. These citations have been removed where they are no longer needed.

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Revision History

Rev	Description	Epistemic Impact	DOI	Date
1.0	Initial release, published by Lex et Ratio Ltd as 'Speaking into Dissonance: Foreign Language Learning, Cognitive Dissonance, and Epistemic Plurality'.	None (baseline release).	10.13140/RG.2.2.22256.93449	2025-09-10
2.0	Substantive revision under series code LXR-2026-EDU-SPKDIS-WP. Central question changed from whether language learning cultivates dissonance tolerance to how explanatory authority, authority over where a difficulty is taken to lie, should be exercised. State and trait claims separated; trait (habituation) argument withdrawn. Epistemic dissonance redefined following Kahl (forthcoming), with its role in the argument specified; agent-relative uncertainty distinguished from evidential underdetermination; four conditions within communicative failure. Sources of difficulty classified by locus and type. Closure, premature closure, foreclosure, delegation and abdication distinguished; formal, practical and epistemic authority distinguished; participation made task-relative. Linguistic, pragmatic and conceptual mismatch distinguished, with thinking-for-speaking and crosslinguistic influence. Distinctiveness claim replaced by a comparative claim. Belief revision distinguished from evidence-sensitive revision. Accent and credibility distinguished from identity-prejudicial testimonial injustice. Brief positioning against accounts of epistemic dependence and authority. Section on AI-mediated language use with a three-part criterion (scope, grounds as justificatory adequacy, developing capacity). Five hypotheses with research designs. Corrections to Version 1.0 in Appendix A. Subtitle changed. Licence changed from CC BY-NC-ND 4.0 to CC BY 4.0.	Substantive: central thesis replaced; principal civic claim of Version 1.0 withdrawn.	10.5281/zenodo.23683678	2026-10-27

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